

PATRICK KACIC

patrick.kacic@yahoo.de | Freiburg im Breisgau, Germany

Data scientist with a focus on environmental data to study ecosystem dynamics, such as disturbances, structure dynamics, and structure-biodiversity relationships.

WORK EXPERIENCE

Postdoctoral researcher at [German Aerospace Center \(DLR\), Munich \(Germany\)](#)

12/2025 - present

- Integrating deep learning algorithms for multi-annual forest structure modeling.
- Sentinel-2, GEDI, and ALS data processing on HPDA cluster.
- Proposal writing, scientific publishing, student supervision, and project administration.

Doctoral researcher at [University of Würzburg \(Germany\)](#)

09/2022 - 11/2025

- Machine learning for multi-annual forest structure modeling using Sentinel-1, Sentinel-2, and GEDI.
- Review on forest biodiversity monitoring based on satellite data.
- Satellite time-series analyses for forest structure monitoring in the context of forest management.
- Ground- and spaceborne-based monitoring of forest structure for biodiversity assessment.
- Proposal writing, scientific publishing, student supervision, field work, and project administration.

Researcher at [German Aerospace Center \(DLR\), Munich \(Germany\)](#)

03/2022 - 08/2022

- Deforestation monitoring in Paraguay using Sentinel-1 and -2 data.
- Organization of training workshop about forest monitoring using satellite data for university researchers, federal employees and NGO workers from Paraguay.
- Project administration, scientific publishing, and student supervision.

Data scientist at [Biocarbon Partners, Lusaka \(Zambia\)](#)

01/2022 - 02/2022

- Deforestation monitoring in Zambia and Mozambique using satellite time-series.
- Forest structure analyses using Sentinel-1, Sentinel-2, and GEDI data.
- Organization of training workshop about forest monitoring using satellite data for researchers.

SELECTED PUBLICATIONS

[Forest structures related to biodiversity can be assessed using remote sensing - Evidence from a large-scale experiment](#)

Kacic et al. 2026, Ecological Informatics (under review).

[Mapping pan-European forest canopy cover loss at monthly resolution using Sentinel-2 and Landsat optical data](#)

Thonfeld, ... Kacic et al. 2026, Remote Sensing of Environment (under review).

[The positive effects of canopy openness across post-disturbance management on insect diversity](#)

Kortmann, ... Kacic et al. 2026, Biological Conservation.

[Characterizing local forest structural complexity based on multi-platform and -sensor derived indicators](#)

Kacic et al. 2025, Ecological Indicators.

[Assessing experimental silvicultural treatments enhancing structural complexity in a central European forest – BEAST time-series analysis based on Sentinel-1 and Sentinel-2](#)

Kacic et al. 2024, Remote Sensing in Ecology and Conservation.

[Forest Structure Characterization in Germany: Novel Products and Analysis Based on GEDI, Sentinel-1 and Sentinel-2 Data](#)

Kacic et al. 2023, Remote Sensing.

[Impacts of national governments on the forest cover loss in Paraguayan Chaco between 1999 and 2021](#)

Salinas, ... Kacic et al. 2023, Journal of Nature Conservation.

[Forest Biodiversity Monitoring Based on Remotely Sensed Spectral Diversity—A Review](#)

Kacic et al. 2022, Remote Sensing.

LANGUAGES

German

Native

English

Fluent (C1)

French

Intermediate (B2)

Italian, Spanish

Basic (A2)

INTERNSHIPS

Master thesis at [German Aerospace Center \(DLR\), Munich \(Germany\)](#)

06/2021 - 12/2021

Deforestation monitoring and forest structure analyses using GEDI, Sentinel-1 and -2.

Student assistant at [University of Freiburg \(Germany\)](#)

01/2020 - 09/2020

Tree growth analyses in the dendrochronology laboratory.

Bachelor thesis at [Eurac Research, Bolzano \(Italy\)](#)

03/2019 - 06/2019

Land cover analyses under alpine disturbances using multi-decadal Landsat time-series.

Intern at [Regional Synergie, Innsbruck \(Austria\)](#)

03/2019 - 06/2019

Online cartography of infrastructure data in the Alps.

Student assistant at [University of Heidelberg \(Germany\)](#)

11/2017 - 08/2018

Laboratory assistant for soil and vegetation analyses, including field work.